

Week 5 Guided Self-Study

- Itanium: Intel's Great Successor
 - YT channel: Asianometry
- Reflect: use the study questions on the next slide
- Focus: register renaming, dynamic scheduling, and memory dependence prediction

Study Questions

1. What scheduling responsibilities does VLIW move from hardware to the compiler, and why can this reduce hardware complexity?
2. Why must trace scheduling respect true data dependencies, and how does compensation code preserve correctness when execution leaves the selected trace?
3. How do repeated loops and predictable control flow help a VLIW compiler find and exploit instruction-level parallelism?
4. Why did AMD64 offer an easier migration path than Itanium, even though both could execute existing 32-bit x86 software?
5. How did continued improvements in x86 processors and the rise of commodity compute clusters weaken the case for adopting Itanium?